

Functional genetic report · longevity

Nitric Oxide and Endothelium

Sérgio F · Report date: 02-10-2026

The endothelium is the inner layer of all vessels; nitric oxide that produces the arteries, prevents the adhesion of platelets and leukocytes and stops atherosclerosis. This report reads the genes of each step of the pathway. Only genes (chip + local imputation + polygenic scores).

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Summary

11

SNPs analyzed

8

in the chip

3

imputed ($DR2 \geq 0.8$)

0

SNP not found in the data

3

Polygenic scores

The nitric oxide pathway has five steps with known genetics: production by endothelial NO- synthase (NOS3), which needs arginine and cofactor BH4 (GCH1); endogenous brake ADMA, eliminated by DDAH1 and AGXT2; the NO receptor, soluble guanylate-cyclase (GUCY1A1), which produces cGMP; degradation of cGMP by PDE5 (the target of tadalafil); and the vasoconstrictor counterweight of endothelin-1 (EDN1). The superoxide of NADPH oxidase (CYBA) destroys the NO. There is also an independent pathway of NOS3: the vegetal nitrate is converted into nitrite by bacteria in the mouth and then into NO.

In synthesis: genetically low ADMA brake (DDAH1 without the allele that increases ADMA) and low coronary risk, but a production of NO by NOS3 slightly below optimal (heterozygous in the variants Glu298Asp and -786T>C) and a copy of the allele that increases endothelin-1. The specific coronary score of the endothelial pathways is at the 52nd percentile (medium). The tadalafil (plus cGMP) and telmisartan (less superoxide via AT1 plus NOS3) that already takes act directly in this pathway.

Favourable

- **DDAH1 rs28489187 AA:** without the G allele that increases ADMA (the endogenous inhibitor of NO-synthase); the strongest locus of ADMA ($p=1e-40$).
- **NOS3 rs3918226 CC:** without the variant of the promoter that reduces NOS3 and increases coronary disease and heart failure.
- **Low coronary risk** (metaGRS at the 5th percentile); **GCH1 GG** (normal synthesis of BH4) and **CYBA GG**.

Unfavourable

- **NOS3 Glu298Asp TG and -786T>C CT:** heterozygous in the two classical variants; minus endothelial NO, small effect.
- **EDN1 Lys198Asn GT:** a copy of the allele that increases endothelin-1 in plasma.
- **GUCY1A1 rs7692387 GG:** homozygous for the common allele of coronary risk (protector A is ~20% of alleles). Coronary score of the endothelial pathways at the 52nd percentile.

Interpretation: The NO pathway is genetically balanced: slightly low production, low brake (ADMA) and low global coronary risk. Diet nitrate pathway and exercise (shear stress) are the ways to increase NO that do not depend on NOS3 variants.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 8 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The PGS005322 score includes only the variants of coronary disease in genes of endothelial cell pathways (36 variants) and is at the 52nd percentile. The coronary metaGRS (1.7 M variants) comes from the complete imputation (the 5th percentile): the global coronary risk is low, and the endothelial component is medium. The VEGF-A score in plasma (protein) is at the 26th percentile. The GWAS Catalog scores for endothelial function and for the NOS3 protein did not have variants with usable effect size and were not used.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Coronary disease: endothelial cell pathways (36 variants)	PGS005322	35 of 36 (97%)	p52	Typical / no judgement
VEGF-A in plasma (protein)	PGS000294	541 615 of 552 070 (98%)	p26	Typical / no judgement
Coronary disease (metaGRS, 1.7 M variants)	PGS000018	1 476 469 of 1 745 179 (85%)	p5	Favourable

Endothelial NO-syntase and NO receptor (NOS3, GUCY1A1)

NOS3 is activated by blood flow shear stress (exercise is the main stimulus), insulin and estrogens. NO diffuses to smooth muscle and activates soluble guanylate-cyclase (GUCY1A1/GUCY1B1), which produces cGMP; cGMP relaxes the artery until it is degraded by PDE5.

Polygenic scores of this theme: Coronary disease: endothelial cell pathways (36 variants) (PGS005322): the 52nd percentile **Typical / no judgement**; Coronary disease (metaGRS, 1.7 M variants) (PGS000018): the 5th percentile **Favourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
NOS3 Glu298Asp	rs1799983	TG	imputation DR2=1.00	T	1	0.35	The Asp298 (T allele) form of endothelial NO- synthase is more sensitive to cleavage and produces less nitric oxide; associated with lower flow-mediated dilation and more stroke. (moderate evidence)	Typical / no judgement
NOS3 -786T>C	rs2070744	CT	chip	C	1	0.44	NOS3 promoter: C allele reduces the transcription in response to the flow (shear stress). (moderate evidence)	Typical / no judgement
NOS3 promoter	rs3918226	CC	chip	T	0	0.09	Variant of the NOS3 promoter (T allele, less common): lower expression, more coronary disease and heart failure. (moderate evidence)	Typical / no judgement
GUCY1A1	rs7692387	GG	chip	G	2	0.80	Soluble guanylate-cyclase, the nitric oxide receptor that produces cGMP: the G allele (common) is associated with more coronary disease, A protects. (moderate evidence)	Typical / no judgement

Favourable

- NOS3 rs3918226 CC
- coronary metaGRS at the 5th percentile

Unfavourable

- NOS3 Glu298Asp TG
- NOS3 -786 CT
- GUCY1A1 rs7692387 GG (common allele)
- Coronary score of the endothelial pathways at the 52nd percentile

Interpretation: Production of NO slightly reduced by NOS3 (two variants in heterozygous), without the low expression allele rs3918226. GUCY1A1 has the most common genotype, which is not the protector. The tadalafil extends the cGMP and partially compensates the two things.

ADMA, the endogenous inhibitor of NO-syntase

ADMA (dimethylarginine asymmetric) competes with arginine and inhibits NOS3; high levels are associated with endothelial dysfunction, coronary disease and mortality. It is eliminated mainly by DDAH1 and, to a lesser extent, by AGXT2 in the kidney.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
DDAH1	rs28489187	AA	chip	G	0	—	DDAH1 degrades ADMA, the endogenous inhibitor of NO-syntase; G allele increases ADMA (p=1e-40). (strong evidence)	Favourable
LOXL4/ PYROXD2	rs10786414	TC	imputation DR2=1.00	T	1	0.66	ADMA locus: C allele low to ADMA; T is the highest ADMA allele. (moderate evidence)	Typical / no judgement
AGXT2 Val140Ile	rs37369	CC	chip	—	—	—	AGXT2 eliminates ADMA and SDMA in the kidney; T allele reduces the enzyme (effect mainly on SDMA). (moderate evidence)	Typical / no judgement

Favourable

- DDAH1 rs28489187 AA (without high ADMA allele)
- AGXT2 rs37369 CC (common form)

Unfavourable

- LOXL4/PYROXD2 rs10786414 TC (a higher allele of ADMA, typical)

Interpretation: Genetically low ADMA lock, which favors the availability of NO. Dapagliflozin and normal renal function help to eliminate ADMA and SDMA.

Endothelin and VEGF

Endothelin-1 is the most potent vasoconstrictor of the endothelium; the balance NO/endothelin defines the tone and inflammation of the wall. VEGF-A stimulates NOS3 and capillaries formation.

Polygenic scores of this theme: VEGF-A in plasma (PGS000294): the 26th percentile Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
EDN1 Lys198Asn	rs53370	GT	chip	T	1	0.22	Endothelin-1, the vasoconstrictor that counterbalances nitric oxide: T allele increases plasma endothelin-1 levels. (moderate evidence)	Typical / no judgement
VEGFA +405G>C	rs2010963	GG	chip	—	—	—	VEGF-A, which stimulates NOS3 and capillaries formation: a common polymorphism of the 5' region. (weak evidence)	Typical / no judgement

Favourable

- VEGFA +405 GG (common form)

Unfavourable

- EDN1 Lys198Asn GT (plus endothelin-1)

Interpretation: A copy of the highest endothelin allele; telmisartan reduces the production of angiotensin II-induced endothelin. VEGF-A in the plasma genetically at the 26th percentile.

BH4 and NADPH oxidase

Without enough BH4, NOS3 "uncouples" and produces superoxide instead of NO. Vascular NADPH oxidase (with p22phox subunit, encoded by CYBA) is the main source of superoxide in the wall and is activated by angiotensin II.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
GCH1	rs841	GG	imputation DR2=1.00	—	—	—	GCH1 produces tetra-hydrobiopterin (BH4), the cofactor without which NOS3 "uncouples" and produces superoxide instead of NO; allele A reduces the inducible BH4. (moderate evidence)	Typical / no judgement
CYBA C242T	rs4673	GG	chip	—	—	—	p22phox of vascular NADPH oxidase (superoxide source that destroys NO); allele A (242T) is associated with less activity. (weak evidence)	Typical / no judgement

Favourable

- GCH1 rs841 GG (BH4 normal inducible)

Unfavourable

- CYBA 242 GG (common form, without the lowest activity variant)

Interpretation: Normal BH4 synthesis. CYBA is common. Telmisartan blocks the activation of NADPH oxidase by angiotensin II.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Vascular Aging	NOS3 Glu298Asp and -786C heterozygous; EDN1 GT	DDAH1 AA; metaGRS p5	Balanced.
Arterial stiffness	NOS3 heterozygous; EDN1 GT	Telmisartan; tadalafil	Compensated by the drugs in use.
Oxidative stress	—	GCH1 GG; Common CYBA	No genetic fragility.
Redox imbalance	—	Normal BH4	Favourable.
Intercellular Communication	GUCY1A1 GG (common allele)	—	Neutral.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Flow exercise	Regular and interval aerobic exercise, more strength training; the shear stress of the flow is the strongest stimulus for the expression and activation of NOS3, and compensates the variants Glu298Asp and -786C. <i>suggestion (not applied)</i>	NOS3 heterozygous
Plant nitrite	Rocket, spinach, beetroot, celery and lettuce every day (about 300-500 mg of nitrate); is the NO pathway that does not depend on NOS3. <i>suggestion (not applied)</i>	NOS3 heterozygous
Oral microbiome	Avoiding oral antiseptics of daily use (chlorhexidine), which eliminates bacteria that reduce nitrate to nitrite; brushing and normal dental thread. <i>suggestion (not applied)</i>	Nitrate-NO pathway
Cocoa polyphenols	Pure cocoa (flavanols) in small amounts increases the flow-mediated dilation in trials; without sugar. <i>suggestion (not applied)</i>	EDN1 GT; NOS3
ADMA and arginine	Not supplementary arginine (without benefit and with damage signal in trials); genetic ADMA is already low. If you want to measure, the ADMA in plasma is an analysis available. <i>suggestion (not applied)</i>	DDAH1 AA
Drugs already in use	The tadalafil (plus cGMP) and telmisartan (less superoxide and endothelin via AT1) act in this pathway; maintain as prescribed. Never associate tadalafil with nitrates. <i>suggestion (not applied)</i>	NO-cGMP
Coordination with the plan	LDL and ApoB as low as possible and high fiber; without alcohol. D3 10 000 IU and omega-3 3.2-4 g/day (already taking fish oil). Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- The NOS3 variants have small and heterogeneous effects between studies; the repetition of the 4 intron (4a/4b) is not read by the chip.
- There is no large GWAS of flow-mediated dilation; the endothelial score is a subset of a coronary score.
- The percentiles compare with the reference population (n=633).
- Only genes: do not use vascular analysis or measurements.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population ($n=633$); effect alleles confirmed by the expected frequency.
- Scores: PGS005322 (coronary disease, endothelial pathways) and PGS000294 (VEGF-A in plasma) calculated in this report; PGS000018 (coronary metaGRS) of the complete imputation.
- Main references: Förstermann 2012 (NO-syntases); Thesaurus 2000 (NOS3 Glu298Asp and cleavage); Nakayama (NOS3 -786T>C); Salvi 2012 (NOS3 rs3918226); Erdmann 2013 (GUCY1A1 and coronary disease); Lüneburg 2014 (GWAS of ADMA, DDAH1); Böger 2009 (ADMA and risk); Lundberg 2008 (via nitrate-nitrite-NO); Kapil 2013 (oral elixir and nitrite).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).